

What is the ‘paradox’ of phase transitions? Is reduction compatible with this paradox?

Introduction

In the context of this debate, ‘paradox’ refers to the reality of phase transitions which occur in finite systems in stark contrast with the theory of phase transitions which requires that they only occur in infinite systems. Unlike classical thermodynamics (CT), Statistical mechanics (SM) is unable to distinguish between phases of matter without resorting to the controversial ‘thermodynamic limit’. If the thermodynamic limit is so controversial that it cannot be accepted, the possibility of the target theory CT reducing to the base theory SM is threatened. It is my view that the possibility of reduction requires the existence of discontinuities in nature and the status of SM as a ‘faithful representation’. Yet the controversy of the thermodynamic limit remains, so it must be argued that it is acceptable with an amendment or asterisk, rather than wholeheartedly in its current state.

The Scope and Structure of this Essay

This essay sets out to present and evaluate the arguments by Bangu and Batterman for and against the existence of discontinuities in nature. Then, Shech on the correspondence relation between the theories and reality and my own contribution in favour of the faithful representation of SM.

Given the two questions of:

1. The existence of discontinuities in nature,
2. The degree to which SM is a faithful representation,

there are four possible positions:

		SM being a Faithful Representation	
		YES	NO
Existence of Discontinuities In Nature	YES	{the position this essay argues for}	{null result}
	NO	{arguably another paradox, beyond the scope of this essay}	{null result}

These two questions must be answered in the affirmative if we are to accept in a meaningful way that reduction is compatible with the paradox of phase transitions. This is because the paradox refers to a contradiction between finite reality and infinite abstract concepts.

Reviewing the table

The column for which SM is not a faithful representation all has null results. This is because we cannot conclude any statement from the supposed abstract system of SM if it has no correspondence relation with nature. Therefore, there will be no paradox regardless of whether we also accept the existence of discontinuities in nature. This is further discussed under the header: ‘Does Statistical Mechanics Faithfully Represent Nature?’

Similarly, to accept SM as a faithful representation and reject the existence of discontinuities in nature (the bottom left quadrant) is a contradiction because a theory which acts as an ontological guide to nature cannot guide us if it makes use of unphysical statements. One might say that there are three tenable positions, but this problem is beyond the scope of this essay.

Overview of the Problem

Statistical Mechanics and the Thermodynamic Limit

First order phase transitions are in their nature instantaneous and distinct – meaning there is a discrete shift from one phase to the other. Both CT and SM must model first order phase transitions by a *discontinuity* to reflect this nature. There are more complex phase transitions of higher orders, but they are beyond the scope of this essay.

Phase Transitions in Thermodynamics

CT predicts phase transitions by the *free energy* in the system – which may be either the Gibbs, G , or Helmholtz, A , type. The only difference between these two quantities is that they correspond to the pressure and the volume, respectively, remaining constant:

$$G = H - TS$$

$$A = U - TS$$

where H is the enthalpy, U is the internal energy, T is the temperature, and S is the entropy.

The discontinuity in the free energy may occur in the free energy or any of its derivatives:

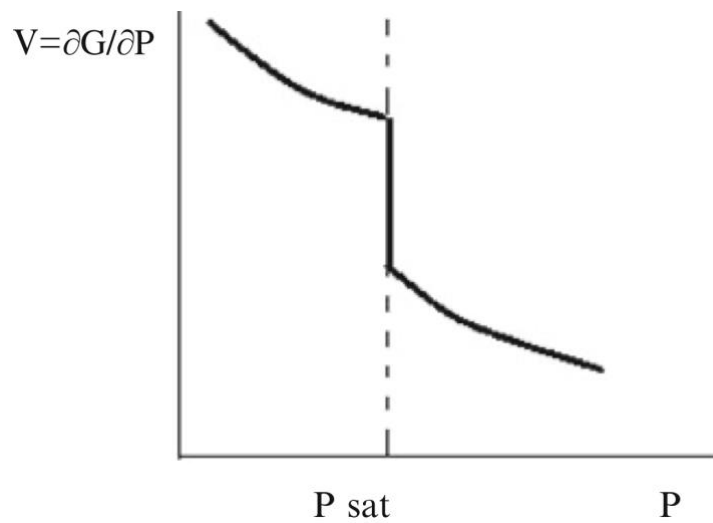


Figure 1 (Bangu, 2015, p.4)

where P_{sat} is the (maximum) pressure at which the gas condenses.

Phase Transitions in Statistical Mechanics

If there is a discontinuity in the free energy, a phase transition must have occurred. SM also models phase transitions by a discontinuity in the free energy given by:

$$A = G = -kT \ln(Z)$$

SM, therefore, needs to find a discontinuity in this equation. Neither the Boltzmann constant, nor the temperature can be, the discontinuity must be present in Z .

Z is the partition function. It is a statistical quantity which sums over all the possible microstates of a system and contains the probability the system being in all its possible states:

$$Z = \sum_i \exp \left(-\frac{E_i}{k_b T} \right)$$

where 'i' is every microstate and E_i is that energy of the 'i'th microstate. The probability of a system being in a certain microstate, i, is weighted by its energy, E_i .

Unfortunately, the partition function is analytic, meaning it is smooth because it has a well-defined derivative at every point. Therefore, a discontinuity must be arrived at by taking the number of particles in the system to infinity – affecting E_i .

The thermodynamic limit: $N \rightarrow \infty$

The discontinuities in the partition function cause discontinuities in the SM free energy as we wanted. Also, thermodynamic quantities can be obtained by taking partial

derivatives of the partition function. Therefore, it is critical to the utility of the partition function that it can achieve a singularity and give a single value.

However, this is a controversial step as it is unphysical, as it is impossible for the system to be infinite. Whether this idealisation is unacceptable is up for debate, but it certainly falls short of our ideal of a fundamental theory that is completely accurate.

Summary

The paradox can be formulated as follows: it seems that an infinite system must be modelled to represent phase transitions in finite systems.

Or, as Shech puts it in his reformulation of Callender's 4 propositions describing the paradox (Shech, 2013, p.4):

1. Concrete systems are composed of finite many particles N ,
2. Concrete systems undergo phase transitions,
3. PT occur if and only if the partition function Z has a discontinuity,
4. The partition function Z of a system with finitely many particles N can only display a discontinuity by appealing to the thermodynamic limit,
5. A system in the thermodynamic limit has infinitely many particles.

To reiterate, there is a paradox of phase transitions if discontinuities exist in nature. But it is only by accepting the existence of discontinuities in nature, and therefore the paradox, that we can begin to argue in favour of the possibility of reduction.

In contrast, there is no paradox of phase transitions if there exist no discontinuities in nature. However, SM is incapable of distinguishing between phases of systems without discontinuities. More importantly, there would be no analogue for phase transitions in SM to those in CT. Without a bridge law, CT cannot be reduced to SM.

The Existence of Discontinuities in Nature

Bangu's argument against belief in discontinuities in nature

One response is to deny the paradox outright as 'it is the attempt to account for a fictional explanandum that demands for a fictional explanans' (Bangu, 2015). Bangu uses a case of an inaccurate idealisation in electrostatics to explain why philosophers are incorrect to insist on a discontinuity in SM.

Bangu asks us to consider a uniformly charged disk of radius R with charge density ϵ_0 . Measuring the electric field at several points both above the disk and along its axis, we find that it is given by:

$$E = \frac{\sigma}{2\epsilon_0}$$

Then, deriving an expression for the electric field, we arrive at:

$$E_H = \frac{\sigma}{2\epsilon_0} [1 - (1 + (R/h)^2)^{-1/2}]$$

Taking the limit $R \rightarrow \infty$, we arrive at the measured value. Furthermore, we can imagine measuring the electric field at two different points and recording identical values, because these two points are so close to each other, that the difference in the electric field falls below the margin of error of our instruments.

Similarly, we do not record the singularity where the discontinuity occurs. It is the result of our representation of the data to have a vertical line connecting two data points instead of a line with a very large gradient. Therefore, the phase transitions observed, with the partial of the free energy being discontinuous, is underdetermined.

Furthermore, it could not possibly be determined by empirical observation because of the inherent limitations of the instruments.

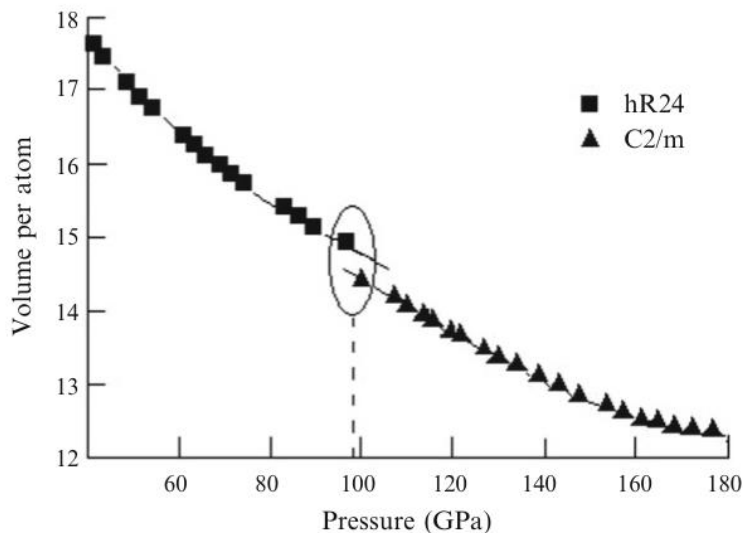


Figure 2 (Bangu, 2015, p.9)

In the image above, the ring encircling the two data points above the dotted line demonstrate the point of discontinuity. If we were to zoom into the diagram, we could always deny the discontinuity by drawing a curve to connect the two datapoints. As the measurements become more accurate however, the curve must become steeper. Yet it remains in possible, in principle, to demonstrate a discontinuity (a completely vertical line).

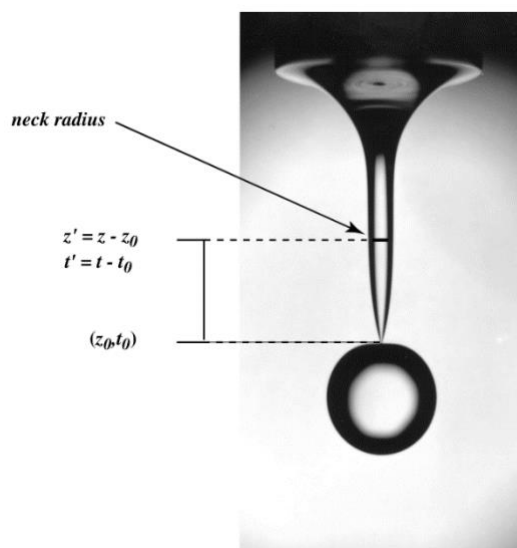
To take the thermodynamic limit to model a phase change is, in light of this consideration, accepting an unphysical statement (the limit itself) to achieve an unphysical result (the unobservable discontinuity). It is worth noting, however, that this argument is not so bold that it rejects the existence of discontinuities in nature, just that our belief in discontinuities is not justified.

If discontinuities are not present in nature, then SM cannot be a faithful representation of nature. Therefore, there is no paradox between the concrete systems not having discontinuities and discontinuities existing in abstractions. In other words, a theory that requires the use of a discontinuities to make predictions cannot be a faithful representation of a world without discontinuities.

Batterman's argument for the existence of discontinuities in nature

Batterman's argument is bolder in that it extends beyond whether our belief in discontinuities may be justified to say that they actually exist. If we can be certain that there are discontinuities in nature, the thermodynamic limit seems far less unphysical than it would if we did not believe in discontinuities at all. The use of the thermodynamic limit would be pardoned as a necessary evil for a faithful representation which produced accurate predictions and, for the most part, accurately described the underlying reality. Like a city map representing the reality of the structure of a city, SM faithfully describes the reality of the system – although imperfectly like the map.

Batterman's argument for the existence of discontinuities in nature is as follows (Batterman, 2004, p.13): The formation and breaking of droplets is governed by gravity and surface tension. At the time of the breaking of the interface of the droplet, the length of the interface continually decreases until it becomes smaller than any other comparable length at the scale of the problem. Surprisingly, the shape of the fluid remains the same such that the size of the fluid differs only by a scale factor at each time. This is a similarity solution. A solution for the size of the droplet, with this geometry in mind, is as follows:



$$h(z', t') = f(t')^\alpha H(\xi)$$

$$\xi = \frac{z'}{f(t')^\beta}$$

Figure 3(Batterman, 2004, p.16-17)

As the length of the interface continues to shrink into the atomic scale, the solution fails. This is because the hydrodynamic theory, from which the solution was derived, did not include a microscopic perspective which is influential at the atomic scale. The discontinuity is real, but hydrodynamic theory fails to account for that reality. This problem further resembles the paradox of phase transitions when we realise that hydrodynamic theory can perfectly predict the geometry of the droplet immediately after and before the interface approaches atomic scales. It is only during the breaking off of the droplet (the phase transition) where the theory fails.

So, it is the decreasing scale of the problem that introduces instability and discontinuity in the system. Rather than conclude emergence, this unexpected behaviour could still be predicted by a more fundamental theory which to which hydrodynamics could be reduced. Batterman's droplet example strongly parallels the

This 'instantaneous failure' is reminiscent of the case Batterman presented in which hydrodynamic theory fails for the range of scales when the droplet boundary becomes microscopic. One objection to this, however, may be that the failure for SM is very specifically defined and that the point of failure for hydrodynamic theory is loosely defined. Therefore, the discontinuity which SM requires does not apply to this argument and therefore we cannot accept the existence of discontinuities in nature. On the contrary, the instantaneous failure for SM, an instantaneous phase transition, is even better defined than the instantaneous failure of a hydrodynamic theory, a loosely defined range distance considered 'microscopic'. Therefore, the discontinuity which SM requires seems more credible.

Besides rendering SM incapable of distinguishing between different phases of a system, rejecting the thermodynamic limit would further threaten the reduction of CT to SM. This is because the presence of a discontinuity provides a very clear bridge law between the two theories. It is precisely the strict definition of a discontinuity which is desirable. To give up this easy connectability (in Nagelian terms) would threaten our hopes of reduction.

Does Statistical Mechanics Faithfully Represent Nature?

Returning to Callender's amended 4-proposition description of the paradox stated in the summary, Shech argues that there is no contradiction between the 5 propositions by appealing to a distinction between 'concrete' phase transitions and mathematical descriptions of them.

In other words, there is no paradox if we accept that finite systems undergo phase transitions and that mathematical representations, abstractions, require the abstract

and unphysical thermodynamic limit to model them – unless we have a correspondence relation between them.

Returning to the table in the introduction of this essay, which outlined the four positions: for there to be a paradox, there must be some correspondence relation between the concrete (the system being described) and the abstract (the theory of SM which describes the system). That is why the column for which SM was not a faithful representation gave no results.

Shech argues that this relation must be justified by an indispensability-type argument – meaning certain mathematical concepts are absolutely necessary in order to understand the world. I offer the No Miracles argument. It may seem less bold than the indispensability-type argument, but it proves more effective than it would seem at a first glance.

I believe that SM is a faithful representation because:

1. It generates predictions from a fundamental, microscopic level. Therefore, it is concerned with, arguably, the ‘most real’ objects.
2. Those predictions are accurate.
3. the theory is powerful, meaning it can generate predictions for a wide variety of systems. The paradox of phase transitions can be considered the exception to the rule, as it can produce predictions for every other application without resorting the controversial thermodynamic limit.

It is by no miracle that SM, in the vast majority of cases, produces accurate predictions from the fundamental objects. It must be truly representing nature.

Accepting the Thermodynamic Limit

Having accepted the existence of discontinuities and the credibility of SM as a guide to the ontology of nature, we address the controversial thermodynamic limit.

The result of the discontinuity may be unphysical, but the observable quantities predicted by SM are accurate. We can employ the ‘No Miracles’ argument again. This time to accept the thermodynamic limit. Meaning, it is by no miracle that the thermodynamic limit is so critical to predictions in SM – something about this limit must be an excellent ‘ontological guide’, as Shech puts it, to reality. This instance of the No Miracles argument is especially convincing as SM only requires the thermodynamic limit in the case of phase transitions and no other scenario – this is the previously discussed instantaneous failure.

The thermodynamic limit remains unphysical and controversial but at least we have a reason to accept it until it can be reformulated. Until then, reduction is only compatible and neither achieved nor disproved.

REFERENCES

Bangu, Sorin (2015). Why does Water Boil? Fictions in Scientific Explanation. In U. Mäki (ed.), *Recent Developments in the Philosophy of Science*. Springer. pp. 319-330.

Shech, Elay (2013). What Is the Paradox of Phase Transitions? *Philosophy of Science* 80 (5):1170-1181.

Batterman, Robert (2004). Critical phenomena and breaking drops: Infinite idealizations in physics. *Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics* 36 (2):225-244.